SUPREMUM AND INFIMUM OF DIMENSIONS OVER A DECENT SPACE ARE ATTAINED AT CLOSED POINTS

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Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Closed points exist in every nonempty decent subset. lemma-dimension-decent-space base (same chapter context, structural). The function u -> dim_x over the reduced closed subspace.
- Finite residue field degree over decent points. lemma-universally-bounded-finite-fibres (same chapter, carded L26058). Boundedness of the degrees of closed points.

Source reference. Original lemma 03K5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2865–2865. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03IL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2878–2878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04N3: focused-seeds/geometry/S9-decent-dimension-decent-space/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0B7I: focused-seeds/geometry/S9-decent-dimension-decent-space/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Dimension as defined in Properties of Spaces, Section 04N3 behaves well on decent algebraic spaces X over S.*

(1) *If $x \in |X|$, then $\dim_x(|X|) = \dim_x(X)$, and*
(2) $\dim(|X|) = \dim(X)$.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Proof of (1). Choose a scheme U with a point $u \in U$ and an étale morphism $h : U \to X$ mapping u to x. By definition the dimension of X at x is $\dim_u(|U|)$. Thus we may pick U such that $\dim_x(X) = \dim(|U|)$. Let d be an integer. If $\dim(U) \geq d$, then there exists a sequence of nontrivial specializations $u_d \rightsquigarrow \ldots \rightsquigarrow u_0$ in U. Taking the image we find a corresponding sequence $h(u_d) \rightsquigarrow \ldots \rightsquigarrow h(u_0)$ each of which is nontrivial by Lemma 03K5. Hence we see that the image of $|U|$ in $|X|$ has dimension at least d. Conversely, suppose that $x_d \rightsquigarrow \ldots \rightsquigarrow x_0$ is a sequence of specializations in $|X|$ with x_0 in the image of $|U| \to |X|$. Then we can lift this to a sequence of specializations in U by Lemma 03IL.

Part (2) is an immediate consequence of part (1), Topology, Lemma 0B7I, and Properties of Spaces, Section 04N3. □